

Tafila Technical University

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Risk Management

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Report-1

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Security Risks of Storing Credit Card Data on an Unencrypted Server

Protecting customer financial data is one of the top priorities for businesses in the digital age. With technological advancements, cyberattacks have become more complex, highlighting the need for stringent security measures. This report addresses the security risks associated with storing customer credit card data on an unencrypted server in a retail company, providing a comprehensive analysis of these risks along with **qualitative** and **quantitative** assessments. The report includes risk classification based on severity, along with the cost and time required to mitigate these risks, and data visualizations for further analysis.

Scenario Description

A retail company relies on an internal server to store customer credit card data. However, this server lacks encryption, making it vulnerable to cyberattacks and breaches.

Identification and Risk Analysis

Risk 1: Data Breach

Actor	Vulnerability	Activity	Qualitative Consequences:	Quantitative Consequences:	Impact	Duration and Cost to Fix
Hackers	Lack of encryption makes data readable if accessed.	Exploiting security flaws to gain access to the data.	Customer data leakage, damage to the company's reputation.	Financial losses estimated in millions of dollars due to compensation and fines.	Critical	**Time to implement:** 2-3 weeks **Cost:** $75,000.

Risk 2: <u>Identity Theft:</u>

Actor	Vulnerability	Activity	Qualitative Consequences:	Quantitative Consequences:	Impact	Duration and Cost to Fix
Cybercriminals or unauthorized insiders.	Lack of multi-factor authentication.	Using stolen information to commit financial fraud.	Legal lawsuits, loss of customer trust.	Regulatory fines estimated in hundreds of thousands of dollars.	High	**Time to implement:** 1-2 weeks. **Cost:** $35,000

Risk 3: <u>Malware Attacks</u>

Actor	Vulnerability	Activity	Qualitative Consequences:	Quantitative Consequences:	Impact	Duration and Cost to Fix
Malicious emails or suspicious downloads.	Lack of up-to-date antivirus software.	Installing malware to gain unauthorized access.	Disruption of operations, leakage of sensitive data.	Financial losses estimated in thousands of dollars per day.	Moderate	**Time to implement:** 1-2 weeks. **Cost:** $20,000

Risk 4: <u>Data Loss Due to Natural or Technical Disasters</u>

Actor	Vulnerability	Activity	Qualitative Consequences:	Quantitative Consequences:	Impact	Duration and Cost to Fix
Natural disasters such as fires or floods.	Lack of a disaster recovery plan.	System failure or hardware damage.	Business operations halted, difficulty in data recovery.	Recovery costs can reach hundreds of thousands of dollars.	Low	**Time to implement:** 3-4 weeks. **Cost:** $50,000

Risk 5: <u>Compliance with Regulations and Standards</u>

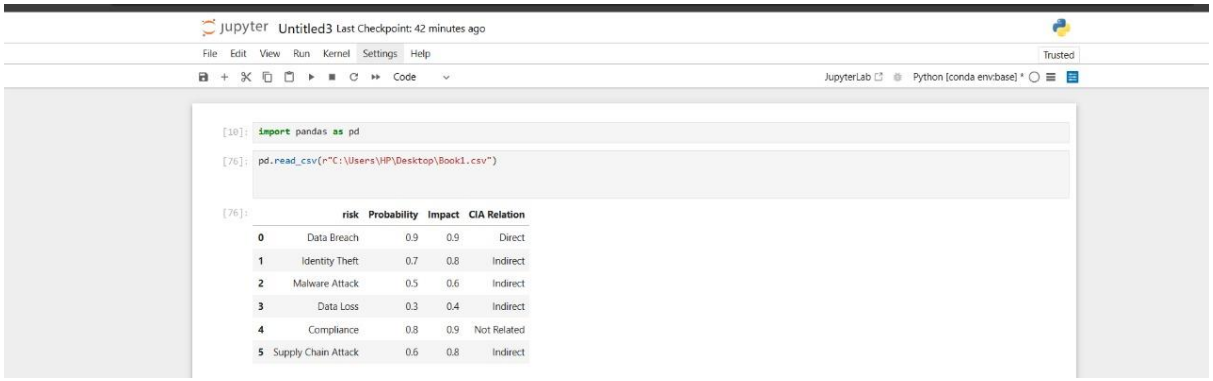
Actor	Vulnerability	Activity	Qualitative Consequences:	Quantitative Consequences:	Impact	Duration and Cost to Fix
Regulatory authorities	Non-compliance with security and privacy standards (e.g., PCI DSS, GDPR).	Audit leading to the discovery of violations.	Fines imposed, business operations suspended.	Fines may reach hundreds of thousands or millions of dollars.	Critical	**Time to implement:** 4-6 weeks. **Cost:** $100,000

Risk 6: <u>Supply Chain Attacks</u>

Actor	Vulnerability	Activity	Qualitative Consequences:	Quantitative Consequences:	Impact	Duration and Cost to Fix
Attackers via partners or suppliers	Weak security among supply chain partners.	Exploiting partners to access company data.	Reputational damage, increased vulnerabilities.	Financial losses could reach millions of dollars.	High	**Time to implement:** 2-3 weeks. **Cost:** $60,000

Charts and Analysis

1. **Virtual dataset**



This virtual dataset highlights the previously mentioned risks, showing their likelihood, impact, and connection to CIA elements. It helps provide a clearer understanding of these risks, their effects on confidentiality, integrity, and availability, and supports finding quick and effective solutions.

2. Analysis of Risks and probability impact

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Code JupyterLab Python [conda env:base] *

[78]:
```
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import pandas as pd

# Sample data
data = {
    'Risk': ['Data Breach', 'Identity Theft', 'Malware Attack', 'Data Loss', 'Compliance', 'Supply Chain Attack'],
    'Probability': [0.9, 0.7, 0.5, 0.3, 0.8, 0.6],
    'Impact': [0.9, 0.8, 0.6, 0.4, 0.9, 0.8]
}

# Create DataFrame
df = pd.DataFrame(data)

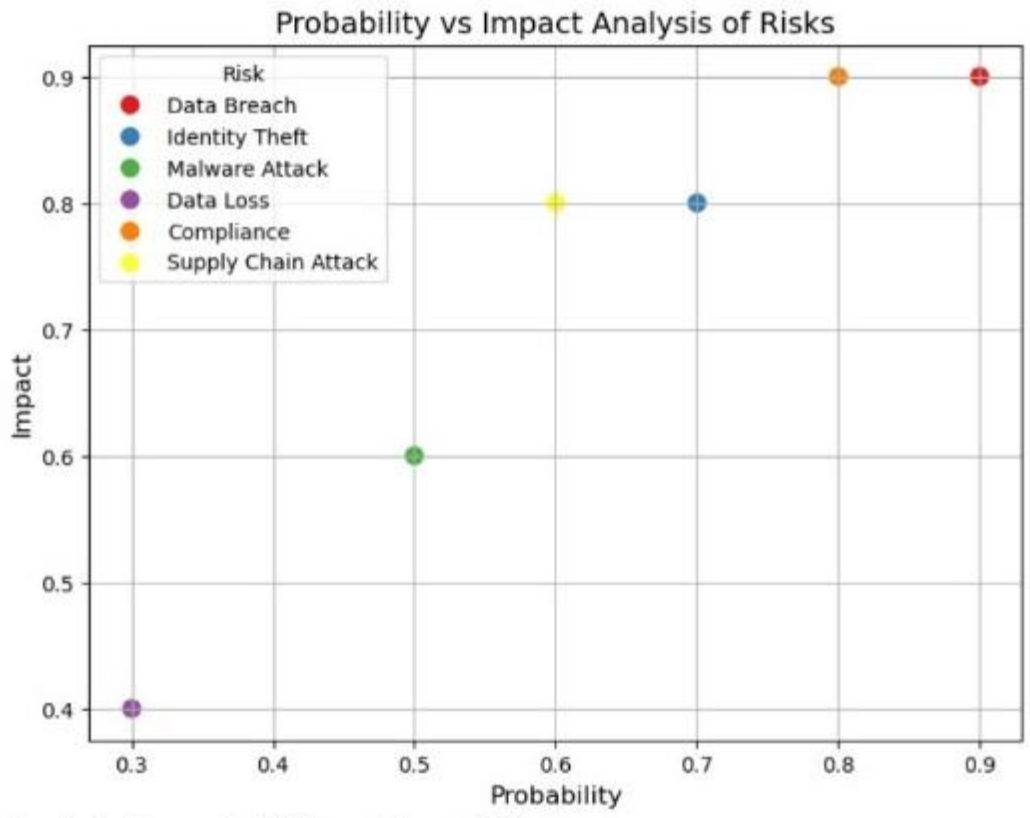
# Create a Scatter Plot between Probability and Impact
plt.figure(figsize=(8, 6))
sns.scatterplot(x='Probability', y='Impact', data=df, s=100, hue='Risk', palette='Set1')

# Add titles and labels
plt.title('Probability vs Impact Analysis of Risks', fontsize=14)
plt.xlabel('Probability', fontsize=12)
plt.ylabel('Impact', fontsize=12)
plt.grid(True)

# Show the plot
plt.show()

# Calculate the correlation coefficient between Probability and Impact
correlation = np.corrcoef(df['Probability'], df['Impact'])[0, 1]
print(f'Correlation between Probability and Impact: {correlation:.2f}')
```

Code




```
Correlation between Probability and Impact: 0.96
```

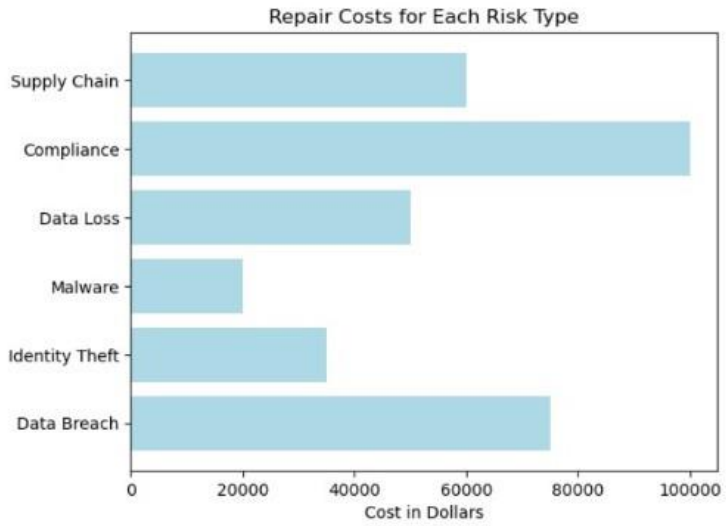
This scatter plot represents the relationship between the probability and impact of each risk. The chart shows that "Data Breach" and "Compliance" present the greatest threats in terms of both probability and impact.

3. Repair Costs for Each Risk Type

```
[80]: types_of_risks = ['Data Breach', 'Identity Theft', 'Malware', 'Data Loss', 'Compliance', 'Supply Chain']

costs = [75000, 35000, 20000, 50000, 100000, 60000]

plt.barh(types_of_risks, costs, color='lightblue')
plt.xlabel('Cost in Dollars')
plt.title('Repair Costs for Each Risk Type')
plt.show()
```

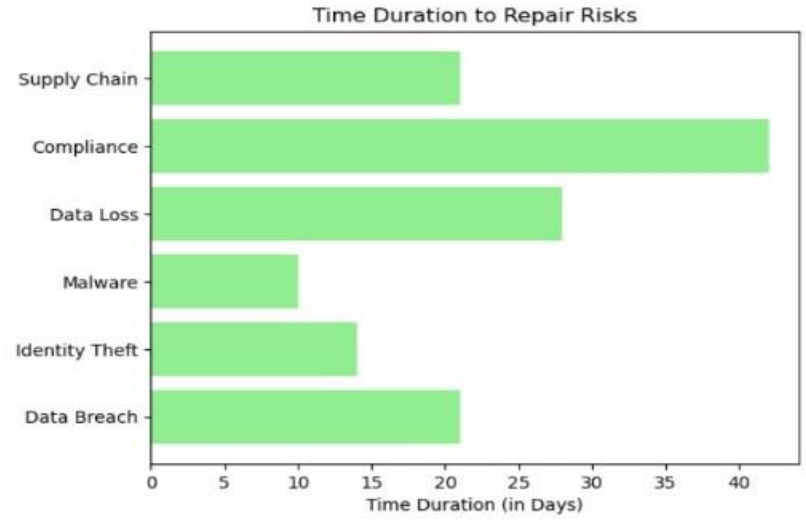


This bar chart displays the **estimated cost** to mitigate each risk, with the highest costs associated with "Data Breach" and "Compliance."

4. Time Duration to Repair Risks

```
[82]: repair_times = [21, 14, 10, 28, 42, 21]

plt.barh(types_of_risks, repair_times, color='lightgreen')
plt.xlabel('Time Duration (in Days)')
plt.title('Time Duration to Repair Risks')
plt.show()
```



This bar chart presents the **expected time** to fix each risk type, showing that "Compliance" requires the longest time to address.

5. CIA Relation to Risks

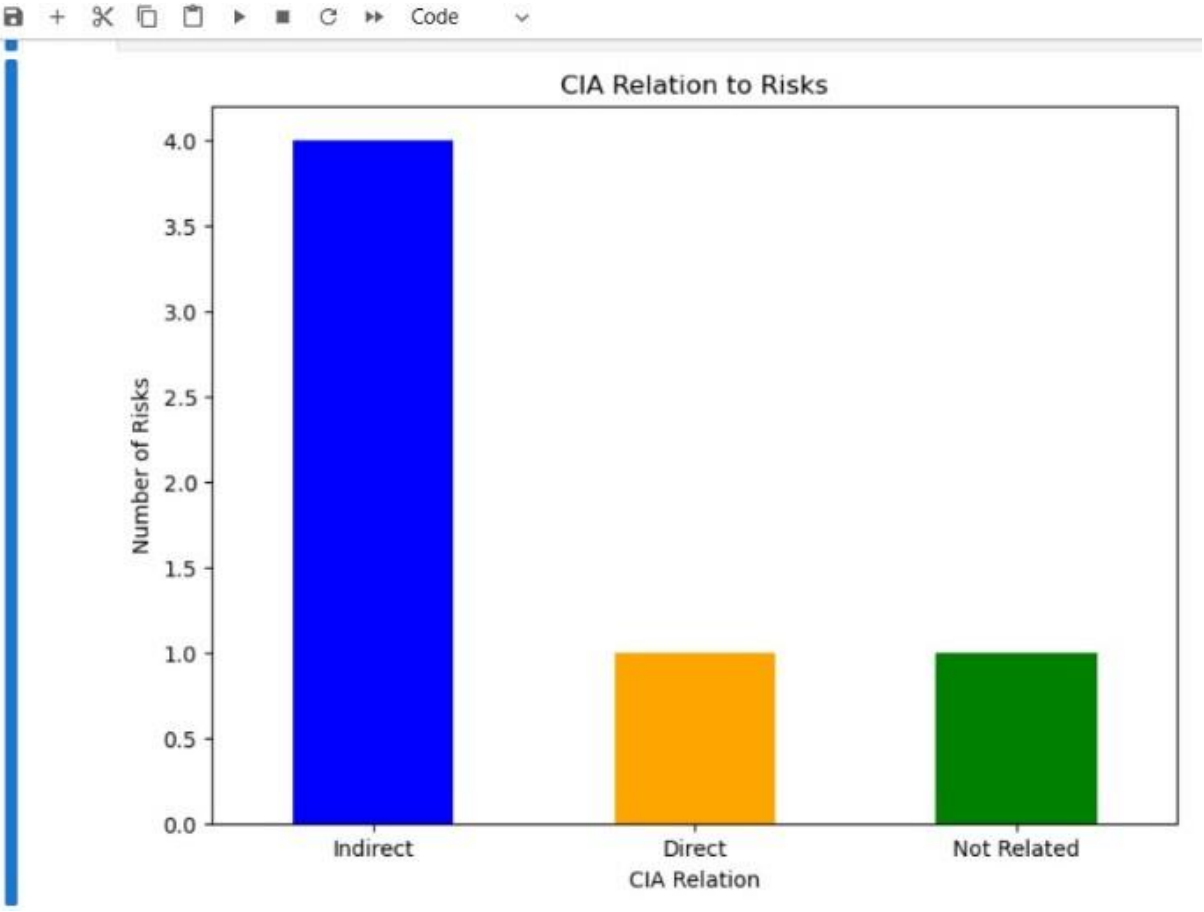
```
[86]: import pandas as pd
      import matplotlib.pyplot as plt

      # Risk data and their CIA relation
      data = {
          'Risk': ['Data Breach', 'Identity Theft', 'Malware Attack', 'Data Loss', 'Compliance', 'Supply Chain Attack'],
          'CIA Relation': ['Direct', 'Indirect', 'Indirect', 'Indirect', 'Not Related', 'Indirect']
      }

      # Create DataFrame
      df = pd.DataFrame(data)

      # Count the number of risks for each CIA relation
      relation_counts = df['CIA Relation'].value_counts()

      # Plot the chart
      plt.figure(figsize=(8,6))
      relation_counts.plot(kind='bar', color=['blue', 'orange', 'green'])
      plt.title('CIA Relation to Risks')
      plt.xlabel('CIA Relation')
      plt.ylabel('Number of Risks')
      plt.xticks(rotation=0)
      plt.show()
```



The bar chart shows the distribution of risks based on their relation to CIA attributes. Most risks are categorized as "Indirect," while a few are "Direct," and one is "Not Related." This highlights how different risks impact the confidentiality, integrity, and availability of data in varying ways.

6. Severity levels of Risk Distribution

```
[100]: import pandas as pd
       import plotly.express as px

       # Risk data with severity levels
       data = {
           'Risk': ['Data Breach', 'Identity Theft', 'Malware Attack', 'Data Loss', 'Compliance', 'Supply Chain Attack'],
           'Severity Level': ['Critical', 'High', 'High', 'Medium', 'Critical', 'High']
       }

       # Create DataFrame
       df = pd.DataFrame(data)

       # Calculate risk distribution by severity level
       severity_counts = df['Severity Level'].value_counts()

       # Interactive pie chart using Plotly
       fig = px.pie(severity_counts,
                    names=severity_counts.index,
                    values=severity_counts.values,
                    title="Risk Distribution by Severity Level",
                    labels={'values': 'Number of Risks', 'names': 'Severity Level'})

       # Show the plot
       fig.show()
```

```
# Interactive pie chart using Plotly
fig = px.pie(severity_counts,
             names=severity_counts.index,
             values=severity_counts.values,
             title="Risk Distribution by Severity Level",
             labels={'values': 'Number of Risks', 'names': 'Severity Level'})

# Show the plot
fig.show()
```



This pie chart illustrates **the distribution** of risks based on severity levels. The results show that over half of the risks fall under the "Critical" and "High" categories, emphasizing the need to focus on these risks

Mitigation Strategies

1. **Data Encryption:** Use strong encryption protocols (AES-256) to secure stored and transmitted data.
2. **Advanced Protection Systems:** Install up-to-date antivirus software and advanced firewalls.
3. **Multi-Factor Authentication (MFA):** Implement two-factor authentication procedures.
4. **Security Awareness and Training:** Train employees to recognize threats and phishing attempts.
5. **Disaster Recovery Plan:** Develop a backup and recovery plan.
6. **Regular Security Audits:** Perform regular security assessments to identify vulnerabilities.
7. **Compliance Assurance:** Review policies to ensure compliance with relevant regulations.
8. **Supply Chain Security Enhancement:** Verify security measures implemented by partners and suppliers.

This report highlights the importance of safeguarding sensitive customer data through comprehensive security measures. Risks can be significantly reduced by employing encryption, enhancing authentication protocols, and developing disaster response plans. Implementing these strategies not only protects data but also boosts customer trust in the company, contributing to its continued success.